

Crosby®

WIRE ROPE END TERMINATIONS

User's Manual

Featuring Application and Installation Procedures for:

National Swage Fittings
Crosby Spelter Sockets
Crosby Mooring Sockets
Crosby Wedge Sockets
Crosby Wire Rope Clips
WIRELOCK® Socketing Resin

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The Market Leader Yesterday, Today and Tomorrow

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General Information

The purpose of this manual is to provide a resource for preparing wire rope terminations when using Crosby products. The publication contains the recommendations of The Crosby Group LLC, OSHA, ASME and the Wire Technical Board. All recommendations in this manual assume the use of Crosby products. Properly following the procedures will produce the most reliable and efficient terminations available. Crosby provides valuable knowledge about swaging, socketing and other termination processes. This knowledge is the result of many years of extensive testing and "on the job" experience.

There are many types of wire rope end terminations that can be produced. Your choice will be dependent upon the application and the task in which the termination is going to be used. The Working Load Limit (WLL) rating of the termination/wire rope combination is not entirely a function of the fitting being used. Therefore, the anticipated efficiency is shown as a function of the wire rope strength. Refer to either the Wire Rope Technical Board's "Wire Rope Sling User's Manual" for the wire rope's rated capacity, or the " Wire Rope User's Manual" for the minimum breaking force of the wire rope, whichever applies to your application.

It is important that the proper components be used to make a termination. For a specific style of wire rope termination you must choose the correct type and size of fitting, the correct swaging dies if a swage product is required, the correct WIRELOCK® Kit if speltered product is required, and proper construction of wire rope as shown in the Crosby General Catalog.

These Crosby instructional materials provide basic application information for the Crosby products and their use with selected other lifting devices. Refer to all manufacturer's warnings, the current Crosby Seminar workbook, the current Crosby General Catalog and current OSHA and ASME standards as they apply.

References to ASME, OSHA and WRTB are for the revisions in effect at time this booklet was printed.

The Crosby Group LLC reserves the right to change product design, materials and specifications without incurring obligations.

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General Cautions and Warnings

All products manufactured by The Crosby Group LLC are sold with the express understanding that the purchaser is thoroughly familiar with the safe and proper use and application of the product.

Responsibility for the use and application of the products rests with the user. The Crosby Group disseminates product warnings and end user application information through various channels. In addition, Crosby provides formal product training seminars and our engineering personnel are readily available to answer your technical questions. For more information read the Crosby General Catalog, refer to Crosby's web site @ www.thecrosbygroup.com, or contact your Crosby Distributor or Crosby direct at 918-834-4611.

Failure of the product can occur due to misapplication, abuse or improper maintenance. Product failure could allow the load to become out of control, resulting in possible property damage, personal injury or death.

There are numerous government and industry standards that cover products made by Crosby. This manual makes no attempt to reference all of them. We do reference the standards that are most frequently asked about.

Ratings shown in Crosby Group literature are applicable only to new or "in as new" condition products.

Load Limit ratings indicate the greatest force or load a product can carry under usual environmental conditions. Shock loading and extraordinary conditions must be taken into account when selecting products for use in a system.

In general, the products displayed in Crosby Group literature are used as parts of a system being employed to accomplish a task. Therefore, we can only recommend within the Working Load Limits, or other stated limitations, the use of products for this purpose.

The Working Load Limit, or Design Factor, or Efficiency Rating of each Crosby product may be affected by wear, misuse, overloading, corrosion, deformation, intentional alteration and other use conditions. Regular inspection must be conducted to determine whether use can be continued at the catalog assigned WLL, a reduced WLL, or whether the product must be withdrawn from service.

Crosby Group products generally are intended for tension or pull. Side loading must be avoided, as it exerts additional force or loading which the product is not designed to accommodate.

Welding Crosby load support parts or products can be hazardous. Knowledge of materials, heat treatment and welding procedures are necessary for proper welding. Crosby Group should be consulted for information.

The assigned Ultimate Load Rating of Crosby Group products for the reeving of wire, manila or synthetic rope is based upon design; the catalog ultimate strength for the rope parts, when totaled, may exceed the assigned Ultimate Load Rating.

The Working Load Limit of a sling must not exceed the lowest Working Load Limit of the components in the system.

The recommended Proof Load on all items in this manual is 2 times the Working Load Limit unless otherwise shown.

Products that Crosby intends for swaging are identified in this manual. For proper swaging machine training, operations and die selection, refer to specific product section in this manual. To develop other product for swaging requires knowledge of materials, heat treatment, product design, die design and performance of the final product.

Use only new genuine Crosby parts as replacements when servicing or repairing Crosby products.

Crosby products are to be considered as sparking, unless otherwise noted.

Product Label Replacement - In accordance with ANSI535.4-1991, "Product Safety Labels" should be periodically inspected and cleaned. "Product Safety Labels" should be replaced when they are no longer legible. Current Crosby warning and application labels, for applicable products, are available from The Crosby Group LLC.

Two decimal and fractional dimensions shown in catalog are intended as nominal dimensions only. If three decimal dimensions are shown, contact Crosby for tolerance information.

Specific warning and application instructions are included in this manual. The instructions can be found at the end of each product section. The symbol shown to the right can be found on the page for products that have application instructions included in this manual. The page numbers that the specific product information can be found are shown in the box for easy reference.



SEE APPLICATION AND WARNING INFORMATION

Para Español: www.thecrosbygroup.com

On Page 00

Low Temperature Service

Crosby forged and cast steel products can be used in general service conditions down to temperatures of -40° F (-40° C). McKissick blocks can be used in general service conditions down to temperatures of -4° F (-20° C). At temperatures from 0° F to -40° F (-18° C to -40° C), good rigging practice requires special attention in the following areas.

1. Lifting should be performed at a steady rate. Shock loading should be avoided.
2. Equipment containing bearings should have increased inspection and maintenance schedule, and may require special lubrication.
3. All lifting equipment should be given a thorough visual inspection before each lift.
4. Remove nicks, gouges, or cracks by grinding (5% maximum material removal).
5. Do not use fittings that have been welded or modified after leaving the factory.
6. If determined to be necessary by the user, lifting equipment should undergo periodic inspection by dye penetrant or magnetic particle surface inspection.

For operation at temperatures below -40° F (-40° C), consider "Cold Tuff" products or contact Crosby Engineering.

Elevated Temperature Service

Crosby forged and cast steel products can be used in general service conditions up to temperatures of 400° F (204° C). The following should be considered when operating up to temperatures of 400° F (204° C).

1. Products that contain non-ferrous materials, and lubricants, plastics, etc. may be adversely affected by high temperatures, and typically should not exceed 200° F (93° C).
2. Galvanized, plated or painted fittings may suffer some or total degradation of the surface finish.
3. Extended exposure to elevated temperatures can cause severe surface scaling and significant permanent reduction of properties.
4. Repeated heating and cooling to room temperatures can result in temper embrittlement.

For other operating temperatures or products, contact Crosby Engineering.

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Definitions

ANSI – American National Standards Institute

ASME – American Society of Mechanical Engineers

ASTM – American Society for Testing and Materials

Boom Pendant – A non-operating rope or strand with end terminations used to support the boom.

Closed Socket – A wire rope end termination consisting of basket and bail made integral (Crosby G-417/S-417 and S-502).

Core – The axial member of a wire rope around which the strands are laid.

Commercial Surface Quality – The surface condition of the products shown in this catalog. The surface condition associated with the normal methods of production of raw materials and machined surfaces. More refined surface qualities are considered as special.

Design Factor – An industry term denoting a product's theoretical reserve capability; usually computed by dividing the catalog Ultimate Load by the Working Load Limit and generally expressed as a ratio, e.g., 5 to 1.

Duplex Non-Tapered Sleeves – Oblong cylindrical sleeves used for fabricating wire rope turnback sling eyes. Also used on grommet slings (Crosby S-506).

Dross – The impurities that form on top of the molten metal.

Extra Improve Plow Steel (EIPS) – A grade of wire rope.

End Termination – The treatment at the end or ends of a length of wire rope, usually made by forming an eye or attaching a fitting and designed to be the permanent end termination on the wire rope that connects it to the load.

Eye or Eye Splice – A loop, with or without a thimble, formed at the end of a wire rope.

Fatique Rated – Tested to a minimum standard of 20,000 cycles at 1.5 times the Working Load Limit. Will meet the requirements of the Euronorm standards for fatique.

Fiber Core (FC) – Cord or rope of natural or synthetic fiber used as the axial member of a strand.

Fist Grip Wire Rope Clips – Fitting for clamping parts of wire rope to each other to form an eye or splice. (Crosby G-429 Fist Grip® Clips)



Flashing – Occurs when sleeve or socket material flows out into the area between the die faces. See picture.

Flemish Eye – A mechanical splice termination formed by unlaying the end of the rope body into two parts, and forming an eye by looping the two parts in opposite directions and laying the rope back together. A swaged metal sleeve secures the ends to the body of the sling.

Full Die – A process where the complete fitting is inserted into the die and swaged following specific swaging procedures for the product.

Full Shank – A process where the full shank length of the fitting is inserted into the die and swaged following specified swaging procedures for the product.

Improved Plow Steel (IPS) – A grade of wire rope. A term for a steel wire cold drawn with a tensile strength of 225,000 to 295,000 P.S.I.

Independent Wire Rope Core (IWRC) – A wire rope used as an axial member of a larger wire rope.

Lang Lay – The type of rope in which the lay of the wires in the strand is in the same direction as the lay of the strand in the rope. The crowns of the wires appear to be at an angle to the axis of the rope.

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Definitions

Lay – a) the manner in which the wires in a strand or the strands in a rope are helically laid.

b) the distance measured parallel to the axis of the rope (or Strand) in which a strand (or wire) makes one complete helical convolution about the core (or center) – In this connection, Lay is also referred to as Lay Length or Pitch.

Lay of Rope –



Marlin Spike – A tapered steel pin used in wire rope splicing.

Open Socket – A wire rope fitting that consists of a basket and two ears with a pin (Crosby G-416/S-416, S-423T, S-501, SB-427).

Plow Steel (PS) – Wire rope steel that contains 0.5 to 0.95 percent carbon.

P.S.I. – Pounds per square inch (Pressure).

Poured Socket – A socket attached by means of zinc or resins.

Progressive Swaging – A process where only a portion (Crosby information is set up on thirds for its swagging machine capacity charts on page 66 in the 2014 Catalog) of the length of the fitting is inserted into the die and swaged following specific swaging procedures for that product.

Proof Load – The average force applied in the performance of a proof test; the average force to which a product may be subjected before deformation occurs.

Proof Test – A test applied to a product solely to determine injurious material or manufacturing defects.

Pull Amount or Tail – The standard amount of wire used in a mechanical splice to complete the splice and give ample gripping area for the sleeve to secure the termination.

QUIC-CHECK® – Swage Sockets incorporate a reduced machined area of the shank end of the swage socket. Before swaging, provides for an obvious visual difference in the shank diameter. After swaging, uniform shank diameter is created allowing for a QUIC CHECK® and permanent visual inspection opportunity.

Designed to quickly determine whether the socket has been through the swaging operation. It does not eliminate the need to perform standard production inspections which include gauging for the proper after swage dimensions or proof loading.

QUIC-PASS® – Crosby swaging process that is completed in just two passes.

Regular Lay – The type of rope where the lay of the wires in the strand is in the opposite direction to the lay of the strand in the rope. The crowns of the wires appear to be parallel to the axis of the rope.

RLL – Right Lang Lay.

Right Regular Lay Rope (RRL) – Strands of wire rope laid to the right with the wire in each strand laid to the left. Right Regular Lay is recommended for use with Crosby National Swage Products.

Seale – The name for a type of strand pattern that has two adjacent layers laid in one operation with any number of uniform sized wires in the outer layer, and with the same number of uniform but smaller sized wires in the inner layer.

Shock Load – A force that results from the rapid application of a force (such as impacting or jerking) or rapid movement of a static load – A shock load significantly adds to the static load.

Socket – Generic name for a type of rope fitting used to terminate wire rope, such as wedge socket, spelter socket, or swage socket (Crosby G-416/S-416, G-417/ S-417, G-517, SB-427, S-421T/S-422T/ S-423T).

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Definitions

Spring Back – A condition that occurs when swaging a fitting to form a termination; whereby the material moves out a small amount when the dies are open.

Swaging – The process of mechanically forming a fitting onto the eye or end of wire rope.

Static Load – The load resulting from a constant applied force or load.

Swaged Fitting – Fitting into which wire rope can be inserted and then permanently attached by swaging the shank that encloses the rope (Crosby S-319SWG, S-409, S-501, S-502, S-505 and S-506).

Thimble – Grooved metal fitting to protect the eye, or fastening loop of a wire rope (Crosby G-408, G-411, G-414, G-414SL, and S-412).

U-Bolt Wire Rope Clips – Fittings for clamping parts of wire rope to each other to form an eye or splice (Crosby G-450 Red-U-Bolt® Clips).

Ultimate Load – The average load or force at which the product fails or no longer supports the load.

Wedge Socket – A wire rope fitting wherein the rope end is secured by a wedge (Crosby S-421T, S-423T and McKissick US-422T Wedge Style Sockets).

Wire Rope Clips – Fitting for clamping parts of wire rope to each other (Crosby G-450 Red-U-Bolt Clips and G-429 Fist Grip® Clips).

WIRELOCK® – A resin based mixture used for forming a 100% efficient termination for wire rope in sockets.

Working Load – The maximum mass or force which the product is authorized to support in a particular service.

Working Load Limit (WLL) – The maximum mass or force which the product is authorized to support in general service when the pull is applied in-line, unless noted otherwise, with respect to the centerline of the product. This term is used interchangeably with the following terms:

1. WLL
2. Rated Load Value
3. Resultant Working Load

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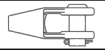
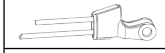
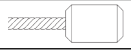
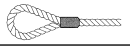
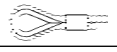
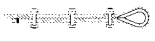
Introduction to Crosby End Terminations

Wire rope is a very useful tool. However, for the wire rope to be utilized, some approved type of termination must be attached. The Crosby Group LLC manufactures a variety of fittings designed specifically to terminate wire rope (see table below). Guidelines and procedures for the application of terminations shown are covered in this manual.

The type of end termination selected is determined by the application and strength required of the final assembly. Each wire rope termination attaches with a special method which may affect the strength of the wire rope.

The terminations and stated efficiency shown in this document are based on the specific wire rope constructions defined for each respective product. It is understood that other constructions of wire rope and strand are available in a variety of different constructions.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope stated in the note for that fitting, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Termination Type		Termination Efficiency
	Swage Sockets	100%
	Spelter Sockets	100%
	Wedge Sockets	80%
	Swage Buttons	98%
	Flemish Eyes	90% or Better
	Turnback Eyes	90% or Better
	Wire Rope Clips	80% or Better

Termination Efficiency

The strength of the wire rope is measured by the concept of efficiency. The efficiency ratings for wire rope end terminations are based upon the minimum breaking force of the wire rope. The table above depicts some of the more common termination types and their efficiencies. Notice the efficiencies range from 80 percent for wedge sockets and wire rope clips to 100 percent for swaged and speltered sockets.

What does efficiency mean in the field? An 80 percent efficient termination will result in only 80 percent of the wire rope's minimum breaking force being available to be utilized. To determine the actual working load limit that can be assigned to a particular termination, the required design factor must then be applied to this reduced breaking strength.

Example #1:

Wire rope clips need to be applied to 5/8" IWRC EIPS wire rope construction. A Crosby 5/8" Wire Rope Clip is considered an 80% efficient termination. The wire rope has a nominal strength of 20.6 tons, and the design factor is 5 to 1. To calculate the allowed WLL for the assembly you would do the following calculations.

$$\text{Ultimate Strength} = \text{Termination Efficiency} \times \text{Rope's Minimum Breaking Force}$$

$$\text{Working Load Limit (WLL)} = \text{Ultimate Strength} \div \text{Design Factor}$$

Or

Ultimate Strength = .80 x 20.6 = 16.5 tons

WLL = 16.5 ÷ 5 = 3.3 tons

In summary, the WLL of the assembly is determined by the efficiency of the termination, the nominal strength of the rope and the design factor.

Example #2:

The WLL needs to be determined for a Flemish Eye mechanical spliced wire rope sling using 5/8" IWRC EIPS wire rope construction. A properly made Flemish Eye sling has an approximate efficiency of 96%. The wire rope has a nominal strength of 20.6 tons, and the design factor is 5 to 1. To calculate the allowed WLL for the assembly you would do the following calculations.

Ultimate Strength = .96 x 20.6 = 19.8 tons

WLL = 19.8 ÷ 5 = 3.9 tons

Note: The efficiencies above are valid only if the terminations are properly fabricated. All ratings given in tons refer to short tons of 2,000 lbs. Ratings given in metric tons equal 2,204 lbs. and are mentioned as "tonnes" (t) or "metric tons".

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Swage Terminations



1500 Ton Hydraulic Swaging Machines

S-501 Open Swage Socket



S-502 Closed Swage Socket



S-319SWG Shank Hook



S-505 Standard Steel Sleeves



S-506 Duplex Sleeve



S-409 Swage Button



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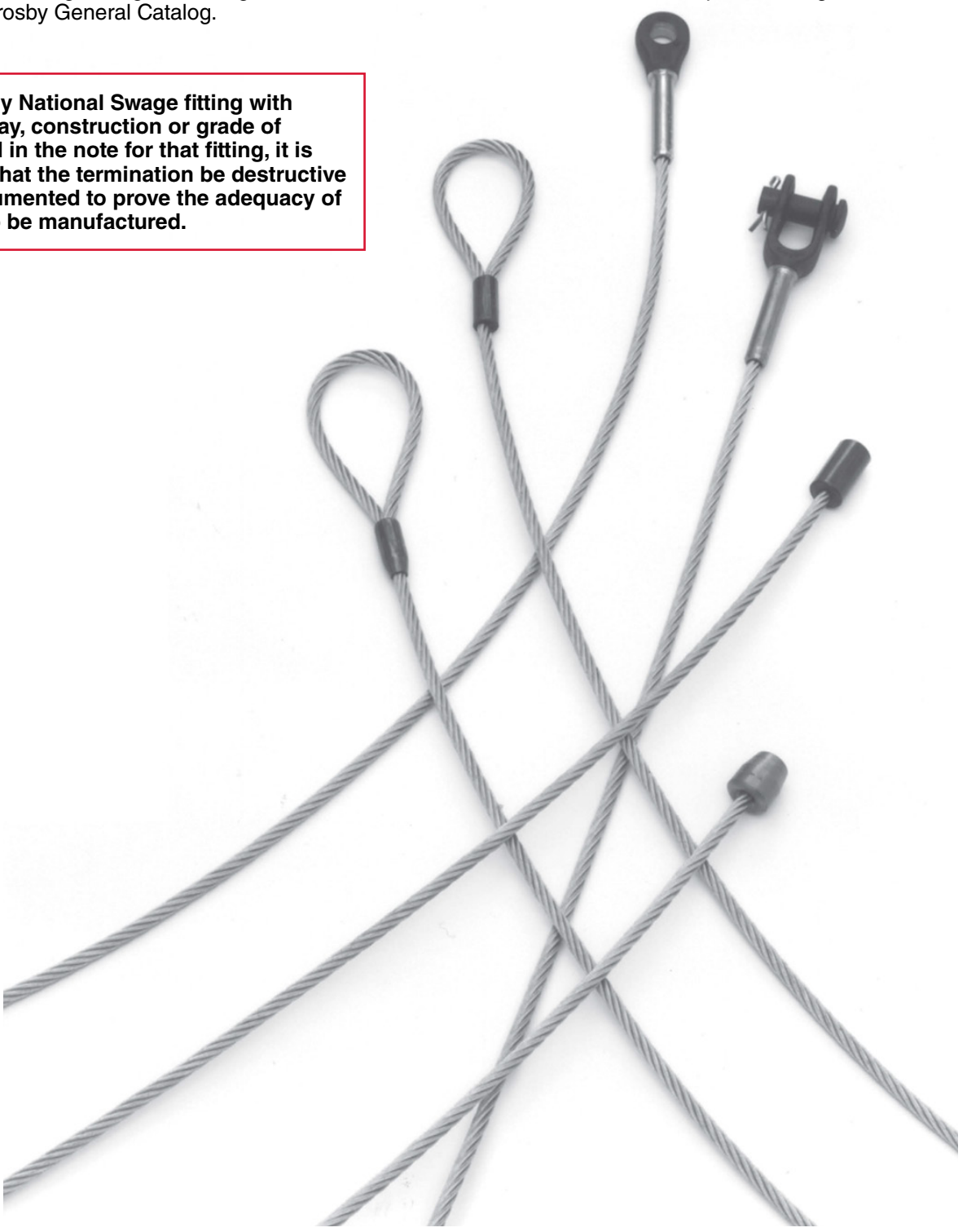
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Introduction to Swaging

There are many styles of terminations that can be made in the end of wire rope using swage fittings. The choice of the style is dependent upon the application and the task. The Working Load Limit rating of the termination/wire rope combination is not entirely a function of the fitting being used. Therefore, the listing of the working load ratings is not within the scope of this literature. (Refer to the Wire Rope Technical Board's "Wire Rope Sling Users Manual" for these values).

It is important that the proper components be used to make a termination. For a specific style wire rope termination, you must choose the correct size and type of fitting and its corresponding swaging die. The information displayed in this manual is based upon using only a one (1) part, 6 X 19 or 6 X 37; FC, IPS or IWRC EIPS, Right Regular Lay Wire Rope. National Swage fittings are designed to be used with inch and metric size wire ropes as designated in this manual and the Crosby General Catalog.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope stated in the note for that fitting, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.



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Swaging Information

The purpose of the swaging process is to produce an efficient termination in the end of wire rope that is permanent. Done properly, the swaging process will produce one of the most reliable and efficient terminations available.

Swaging is a cold forming fabrication method used to permanently attach a specially heat treated forged steel fitting to wire rope in order to form slings, boom pendants or other useful devices for lifting.

The hydraulic swaging equipment, required to provide the slow speed and high pressure necessary for ideal swaging control, causes considerable movement of the steel fitting. The extreme pressure forces the fitting's steel to flow into the crevices between the wires and strands of the wire rope, and at the same time forces the flow along the length of the wire rope.

There are many types of swaged fittings, including:
- Sleeves
- Sockets
- Shank hooks
- Threaded studs

Efficiency

Efficiency ratings for wire rope end terminations are based upon the minimum breaking force of wire rope.

Termination efficiencies for swaged fittings typically range from 90 to 100 percent.

The efficiency of the termination depends on:
1. The type of fitting
2. Utilizing proper swaging techniques
3. Type and construction of wire rope

These will be covered in greater detail in this section of the manual.

This publication contains the recommendations of The Crosby Group LLC for the use of its products **to produce such terminations.**

National Swage, a member of The Crosby Group LLC, provides the highest quality fittings and equipment available for fabricating swaged wire rope terminations. In addition, National Swage also provides valuable knowledge about the swaging process that is a result of the combination of many years of extensive testing and "on-the-job" training.

After Swage Dimensions

One of the important considerations in producing a quality termination is the overall diameter of the fitting after the swaging process is complete. Since all dies wear, and the swaged fitting used in terminations have spring back, the results of swaging should be checked after each swaging for proper after swage dimensions to ensure proper sling performance and to determine the wear condition of the die (Per the Warning on page 71 of the 2014 General Catalog. Third statement). In addition to worn dies, not achieving the proper after swage dimension can also be due to the die not being fully closed during swaging. Dies showing excessive wear should be replaced.

The effective swaging that dies can accomplish stops when the die lands touch each other. Any continued swaging adds needless wear and strain on the dies and swaging machine. By placing a light oil on the die faces and in the cavity, the dies will be lubricated as well as protected. The oozing of the oil from the faces of the dies as they touch will indicate when the dies have closed. At this point, stop the swaging cycle. Again, additional swaging adds needless wear and strain to the dies and swaging machine.

For the proper after swage dimensions, see the section in this publication or the General Catalog for the specific product you are swaging.

Key Facts About Swaging:

1. There are a many styles of terminations that can be made in the end of wire rope using swage fittings. The choice of the style is dependent upon the application and the task.
2. The Working Load Limit rating of the termination/ wire rope combination is not entirely a function of the fitting being used. Therefore, the listing of the working load ratings is not within the scope of this literature. (Refer to the Wire Rope Technical Board's "Wire Rope Sling User's Manual" for these values).
3. It is important that the proper components be used to make a termination. For a specific style wire rope termination, you must choose the correct size and type of fitting and its corresponding swaging die.
4. The information displayed in this manual is based upon using only a one (1) part, 6 X 19 or 6 X 37; FC, IPS or IWRC EIPS, Right Regular Lay Wire Rope.
5. National Swage fittings are designed to be used with inch and metric size wire ropes as designated in this manual and the Crosby General Catalog.
6. Before using any National Swage fitting with any other type lay, construction or grade of wire rope stated in the note for this fitting, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

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National Hydraulic Swaging Machines

SEE APPLICATION AND WARNING INFORMATION
On Pages 11-18

National offers the highest quality and most complete line of Hydraulic Swaging Machines specifically designed to be used to swage fittings on wire rope.

3 Capacities to Fit your Requirements:

- **500 Tons**
- **1000 Tons**
- **1500 Tons**



1500 Ton Hydraulic Swaging Machines
Approximate weight 29,486 lbs. Overall height 8 ft. 10 in.
Fast advance and retract speed.
Automatic slow, precision swaging speed.

The following features of National Hydraulic Swaging Machines offer a number of advantages for high production sling shops:

- ✓ A dual hydraulic system which combines the use of high speed and low pressure to bring dies into position; and the low speed and high pressure necessary for ideal swaging control.
- ✓ Adjustable tonnage control, so tonnage can be set to match die block Working Load Limit.
- ✓ Four column wide stance which provides the operator ample working clearance between columns and a large area for in-process sling storage.
- ✓ Vertical swage action which gives an equalized press on the fitting to produce uniform high quality.
- ✓ Self locating spring locks snap the shoe-type dies into place for quick set-up and change.
- ✓ The National four column wide stance Hydraulic Swaging Machines, each equipped with an up-acting ram, have two side cylinders for fast approach and return of the main ram. They come in three swaging capacities.
  - 500 tons (4450 kN)
  - 1000 tons (8900 kN)
  - 1500 tons (13350 kN)

Swaging Machine Capacity Chart for Standard Round Sleeves and Buttons

Hydraulic Swaging Machine Size	Swaging Method	Die Size (in.)	Largest Fitting Allowed to be Swaged (in.)		
			S-505 Sleeve	S-506 Sleeve	S-409 Buttons
500 Ton	Full Die	2-1/2 x 5 Mark Series 4 x 7 5 x 7	1-1/2	1-1/4*	7/8
1000 Ton	Full Die	4 x 7 5 x 7	2-1/2	1-1/4*	1-1/4*
1500 Ton	Full Die	5 x 7 6 x 12	3-1/2	1-1/4*	1-1/4*

* Largest size fitting available.

NOTE: For special applications or conditions, contact Crosby National at (501) 982-3112.

Swaging Machine Capacity Chart for S-501 and S-502 Swage Sockets

Hydraulic Swaging Machine Size	Swaging Method	Die Size (in.)	Largest Fitting Allowed to be Swaged (in.)
500 Ton	Full Shank	2-1/2 x 5 Mark Series 4 x 7 5 x 7	3/4
	Progressive	4 x 7 5 x 7	1-1/4
1000 Ton	Full Shank	4 x 7 5 x 7	1
	Progressive	4 x 7 5 x 7	1-1/2
1500 Ton	Full Shank	5 x 7 6 x 12	1-1/4
	Progressive	5 x 7 6 x 12	2*

* Largest size fitting available.

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NATIONAL SWAGING MACHINE AND DIE WARNING, USE, MAINTENANCE AND APPLICATION INFORMATION

National Four Post Swaging Machine



⚠ WARNING

- **Misuse of swaging machine can result in serious injury or death.**
- **READ, UNDERSTAND, AND FOLLOW all the information in this warning document and the instructions shown in "Wire Rope End Terminations User's Manual" before operating the swaging machine.**
- **Swaging machine operators must be trained in accordance with the information supplied by The Crosby Group LLC. THE SWAGING MACHINE OWNER IS RESPONSIBLE FOR THE TRAINING AND THE SAFE OPERATION OF THE SWAGING MACHINE.**
- **Do not swage oversize parts.**
- **Only swage parts of the proper design, material and hardness.**
- **If misused, dies and/or die holders may break. PROTECT YOURSELF AND OTHERS:Always stay away from the sides of the swaging machine during swaging operations and alert others in your work area.**
- **Do not shim between dies.**
- **Do not shim die or die holder unless swaging aluminum sleeves**
- **Do not use die holders that are damaged or have loose side rails or side plates.**
- **Keep head, hands, and body away from moving swaging machine and die parts.**
- **Consult die manufacturer for correct use of their product.**
- **Adjust swaging machine tonnage to the Working Load Limit (WLL) tonnage shown on the die block being used. If the Working Load Limit is not legible, refer to Die height & width and corresponding Working Load Limit (See Table 1). Failure to do so can result in serious injury or death.**

Operation Safety

- NEVER use dies that are cracked, worn or abraded (galled).
- NEVER use dies that have an oversized cavity.
- ALWAYS use a matched set of dies.
- When swaging steel fittings,DO NOT SHIM DIES. Dies for steel fittings must be free to float and align one to the other.
- When swaging aluminum fittings, THE STEEL DIES MUST BE SHIMMED. Shim the side of the die to ensure the proper cavity alignment for flash removal.
- NEVER shim between the dies.
- When Swaging Crosby National fittings, use only the proper capacity swaging machine for the size of fitting used (See Swaging Capacity Chart). If the swaging machine capacity exceeds the die block Working Load Limit rating, adjust the swaging machine tonnage to the Working Load Limit shown on the die block being used. See Table 1 for die block Working Load Limit.
- Always use the correct size and type of die for the size wire rope fitting used.
- Always lubricate die faces and cavities with light weight oil.
- Progressive swaging of fittings must be done in accordance with procedure shown in *"Wire Rope End Terminations User's Manual"*. Only open channel dies are to be used.
- Stop swaging when the cavity side of both dies touch.
- Make sure part is swaged to the recommended after swage dimensions (See Crosby General Catalog or *"Wire Rope End Terminations User's Manual"*, Die Guide, or Die Chart).
- If a swage fitting other than a Crosby National is used, determine adequacy of the termination by a destructive pull test.
- All swage sockets must be swaged with socket head adjacent to the socket relief (largest radius) on the die.
- For special applications or conditions, contact Crosby National.

TABLE 1

Die Size (Height x Width)	Working Load Limit (WLL)*
2" x 3-1/2"	200 Ton Mark Series
2-1/2" x 4"	200 Ton National
2-1/2" x 5"	500 Ton Mark Series
4" x 7"	1,200 Ton Mark Series
5" x 7"	1,500 Ton National
6" x 12"	3,000 Ton National

* Note: These Working Load Limits are for Crosby® National Die Blocks only. The Working Load Limits of die blocks from other manufacturers may vary.

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Inspection Maintenance Safety

- Make sure all bolts and nuts are in place and tightened to recommended torque as shown in Table A, on page 13 for new style swaging machines, and Table B on page 14 for legacy swaging machines.
- Load block or die base plate surfaces must be to manufacturers specifications for thickness and flatness to provide complete support of the die during swaging.
- Make sure die holder side rails are not bent, loose or damaged.
- Clean dies and die holder surfaces. Keep free of metal shavings, slag, grit, sand, floor dry, etc.
- Lubricate the four guide bushings daily with light oil.
- Inspect the rods for corrosion. Use #000 emery cloth or steel wool to maintain a high polish surface.
- Do not increase the hydraulic system pressure above the factory preset pressure of: 6500 psi for 500 ton, 1000 ton and 1500 ton swaging machines – 5000 psi for 3000 ton swaging machine.
- Under ordinary operating conditions, drain and clean reservoir every two (2) years.
- Filters inside of the reservoir should be cleaned every time the reservoir is drained and cleaned. The Racine "tell-tale" suction filter should be cleaned every six (6) months.

Die Working Load Limit Pressure Adjustment on Lower Cylinder National 500 Ton through 1500 Ton Swaging Machines

Follow this procedure to adjust swaging tonnage (pressure) on your swaging machine.

1. Install the die holder(s) or die adapter with the dies to be used.
2. Bring the dies together (without a part in the dies) until they just touch.
3. Turn the tonnage control valve, which is located on the control panel left of the tonnage gauge, counter-clockwise about (6) six turns or until knob no longer turns.
4. Now (without a part in the dies) apply pressure to the dies by pressing the foot pedal marked "up".
   A. If the tonnage is lower than desired Working Load Limit, turn the valve clockwise while continuing to press the foot pedal marked "up" until desired Working Load Limit is reached.
   B. If tonnage is higher than desired Working Load Limit, release pressure by pressing the pedal marked "down". Then repeat steps 2 through 4.

Die Working Load Limit Pressure Adjustment on 3000 Ton Swaging Machine

For reducing tonnage, use selector switch on front of control panel to select lower tonnage (approximately 1500 Tons) or 3000 Ton.

WARNING ALWAYS USE 5 X 7 OR 6 X 12 DIES AT 1500 TON SETTING.

WARNING USE ONLY 6 X 12 DIES ON TONNAGE THAT EXCEEDS 1500 TONS.

Swaging Machine Capacity Chart for Standard Round Sleeves and Buttons

Hydraulic Swaging Machine Size	Swaging Method	Die Size (in.)	Largest Fitting Allowed to be Swaged (in.)		
			S-505 Sleeve	S-506 Sleeve	S-409 Buttons
500 Ton	Full Die	Mark Series 2-1/2 x 5 4 x 5 5 x 7	1-1/2	1-1/4*	7/8
1000 Ton	Full Die	4 x 7 5 x 7	2-1/2	1-1/4*	1-1/4*
1500 Ton	Full Die	5 x 7 6 x 12	3-1/2	1-1/4*	1-1/4*
3000 Ton	Full Die	6 x 12	4-1/2	1-1/4*	1-1/4*

* Largest size fitting available.

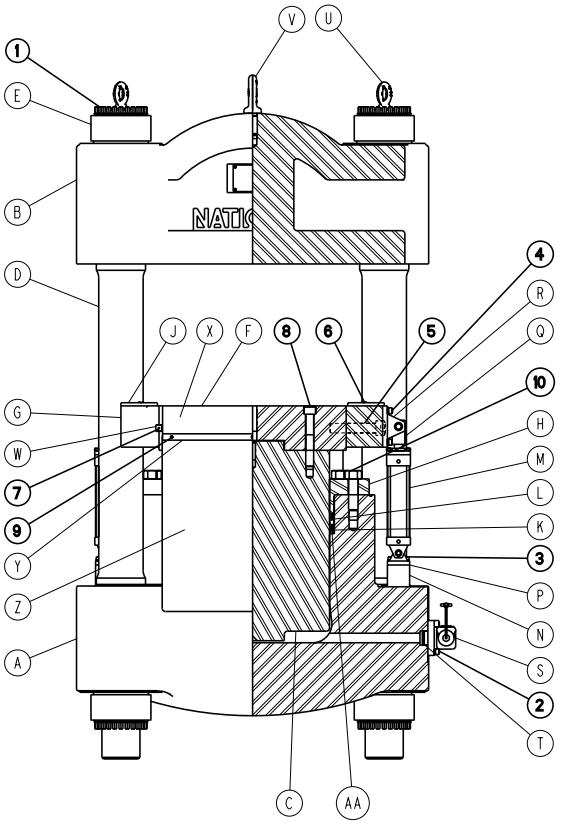
Swaging Machine Capacity Chart for S-501 and S-502 Swage Socket

Hydraulic Swaging Machine Size	Swaging Method	Die Size (in.)	Largest Fitting Allowed to be Swaged (in.)
500 Tons	Full Shank	Mark Series 2-1/2 x 5 4 x 7 5 x 7	3/4
	Progressive	4 x 7 5 x 7	1-1/4
1000 Tons	Full Shank	4 x 7 5 x 7	1
	Progressive	4 x 7	1-1/2
1500 Tons	Full Shank	5 x 7 6 x 12	1-1/4
	Progressive	5 x 7 6 x 12	2
3000 Tons	Full Shank	6 x 12	2*
	Progressive	6 x 12	2-1/2

* Largest size fitting available.

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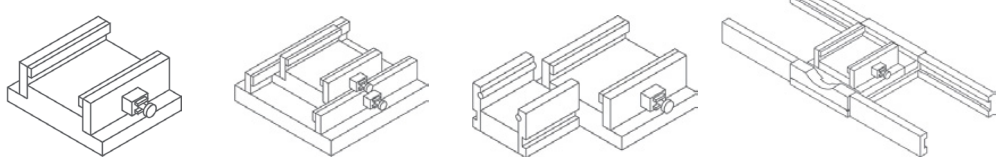
NATIONAL HYDRAULIC SWAGING MACHINE
TORQUE MAINTENANCE INFORMATION



Item	No. Req'd.	Description
A	1	Cylinder
B	1	Housing Cap
C	1	Piston
D	4	Tie Rod
E	8	Tie Rod Nut
F	1	Platen
G	2	Guide
H	1	Gland
J	4	Bushing
K	1	Mono Seal
L	1	Seal Spacer
M	2	Side Cylinder
N	2	Side Cylinder Mount
P	2	Lower Bracket
Q	2	Knuckle
R	2	Upper Bracket
S	1	Check Valve
T	1	Check Valve Seal
U	4	Tie Rod Eyebolt
V	1	Cap Eyebolt
W	4	Key
X	2	Bumper
Y	2	Bumper Strip
Z	2	Rubber Skirt
AA	1	Bottom of Seal Cavity

Table B

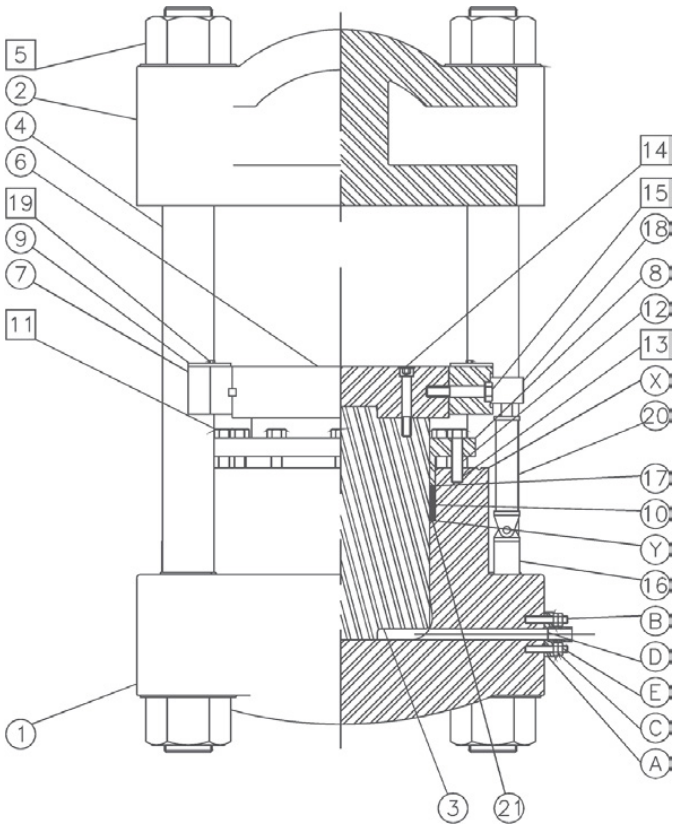
Item No.	No. Req'd.	Description	Torque in Ft-Lbs			
			500 Ton Swaging Machine	1000 Ton Swaging Machine	1500 Ton Swaging Machine	Maintenance Schedule
1	Varies	Tie Rod Nut Jack-Bolts	105	N/A	260	Weekly
2	4	Check Valve Bolts	100	100	100	Weekly
3	8	Lower Bracket Bolts	100	100	100	Weekly
4	8	Upper Bracket Bolts	100	100	100	Weekly
5	4	Guide Bolts	250	250	250	Weekly
6	8	Bushing Screws	15	15	15	Weekly
7	4	Key Screws	4	4	4	Weekly
8	4	Platen Bolts	525	600	700	Monthly
9	6	Bumper Screws	4	4	4	Monthly
10	12	Gland Bolts	700	800	800	Monthly



	Torque in ft./ lbs.
Die Holder Bolt Torque	1/4 - 20 UNC 13
	5/16 - 18 UNC 15
	5/8 - 11 UNC 211
	7/8 - 9 UNC 583

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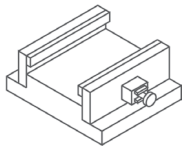
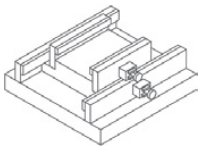
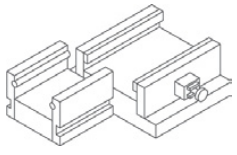
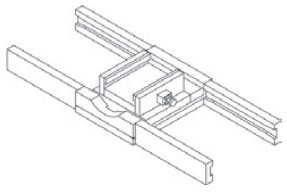
NATIONAL LEGACY HYDRAULIC SWAGING MACHINE
TORQUE MAINTENANCE INFORMATION



Item	No. Req'd.	Description
1	1	Cylinder
2	1	Housing Cap
3	1	Piston
4	4	Tie Rod
5	8	Nut
6	1	Platen
7	2	Guide
8	1	Gland
9	4	Bushing
10	1	Packing Set
11	12	Packing Gland Nut
12	4	Packing Gland Spacer
13	12	Stud
14	4	Cap Screw
15	4	Cap Screw
16	2	Lower Bracket
17	1	Upper Bronze Ring
18	2	Upper Bracket
19	8	Machine Screw
20	2	Side Cylinder
21	1	Lower Bronze Ring
A	1	Block
B	4	Stud
C	4	Nut
D	1	Copper "O" Ring
E	4	Lock Nut
X	1	Top of Cylinder
Y	1	Bottom of Packing Cavity

Table A

Item No.	Description	Torque in Ft. Lbs. 500 Ton Swaging Machine	800 Ton Swaging Machine	1000 Ton Swaging Machine	1500 Ton Swaging Machine	Maintenance Schedule
5	Tie Rod Nuts	2000	2250	2500	2500	Weekly
14	Piston Bolts	525	600	600	700	Monthly
11	Packing Gland Nuts (over spacers only) "all others hand tighten"	200	200	200	200	Weekly
15	Platen Guide Bolts	250	250	250	250	Weekly
13	Packing Gland Bolts	700	800	800	800	6 Months
	Side Cylinder Bolts	100	N/A	100	150	Weekly
19	Guide Bushing Bolts	15	15	15	15	Weekly
	80 M Piston Pump Pistons	96 to 125 all Swaging Machines				

	Torque in ft./ lbs.
Die Holder	1/4 - 20 UNC 13
Bolt Torque	5/16 - 18 UNC 15
	5/8 - 11 UNC 211
	7/8 - 9 UNC 583

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Die Information

CAUTION ⚠

- **Improper die selection could result in significant loss of efficiency in the termination.**

National dies and die holders are made solely for swaging properly designed fittings on wire rope, and any other uses are prohibited.

The swaging operation results in a high degree of cold metal flow. The movement that occurs between the fitting and the dies will cause wear of the dies. Therefore, to prolong the life of the dies, it is important to always lubricate die faces and cavities between each pass with a light weight oil or high pressure grease.

When scores appear in the die cavities, the dies should be removed from service.

NEVER EXCEED THE WORKING LOAD LIMIT OF DIES OR DIE HOLDERS.

All National Standard dies 1/4" through 1" include an open channel die cavity and a tapered die cavity in the same die block.

Dies for S-505 Standard Steel Sleeves (Flemish Eyes)

Die sizes for 1/4" through 1"

Swaging 1/4" through 1" Standard Steel S-505 sleeves on Flemish Eye terminations requires the use of the taper cavity only. Refer to page 24 of the *Wire Rope End Termination User's Manual* for proper die selection.

Die sizes for 1-1/8" and above

Swaging 1-1/8" and larger Standard Steel S-505 sleeves on Flemish Eye terminations requires using 2 sets of open channel dies (1st stage and 2nd stage) for each size. Beginning with the 1st stage die and finishing with the 2nd stage die will achieve proper after swage dimensions. Dies for S-505 Sleeves 1-1/8" and larger are single cavity with open channel. Refer to page 24 of the *Wire Rope End Termination User's Manual* for proper die selection.

Using S-505 Sleeves with Metric Ropes

Although Crosby National S-505 Standard Steel sleeves are designed to be used with most metric ropes, there are selected "intermediate" sizes of metric ropes that when swaged in standard National dies utilizing Crosby National S-505 sleeves do not achieve required after swage dimensions and efficiencies. To ensure all S-505 sleeves achieve the required efficiency when used with metric ropes, Crosby provides special National swaging dies to be used in conjunction with selected size metric ropes. These new dies will produce the required efficiencies and after swage dimensions.

The table found on Page 24 of the *Wire Rope End Termination User's Manual* identifies the new dies that are required to properly swage the selected intermediate size wire ropes not covered in the standard product offering found on Page 25 of the manual.

Dies for 6mm through 26mm (except 12mm, 20mm and 24mm)

Swaging on 6mm through 26mm metric ropes for Flemish Eye slings requires the selection of the proper S-505 Standard Steel sleeve and the use of the tapered cavity only. Refer to page 24 of the *Wire Rope End Termination User's Manual* for proper sleeve and die selection.

Dies for 12mm, 20mm and 24mm

Swaging on 12mm, 20mm and 24mm metric ropes for Flemish Eye slings requires the selection of the proper S-505 Standard Steel sleeve and the use of both the open cavity and tapered cavity in special dies. Refer to page 25 of the *Wire Rope End Termination User's Manual* for proper sleeve and die selection.

Dies for 28mm and larger

Swaging on 28mm and larger metric ropes for Flemish Eye slings requires the selection of the proper S-505 Standard Steel sleeve and the use of 2 sets of open channel dies (1st stage and 2nd stage) for each size. Beginning with the 1st stage die and finishing with the 2nd stage die will achieve proper after swage dimensions. Dies for S-505 sleeves 28mm and larger are single cavity with open channel. Refer to page 24 and 25 of the *Wire Rope End Termination User's Manual* for proper sleeve and die selection.

Important: If the specific size metric rope required is not listed on page 24 of the *Wire Rope End Termination User's Manual* refer to Intermediate Metric Die Chart on page 25 of the manual for proper sleeve and die selection.

Dies for QUIC-PASS® Swaging System – 1/4" through 1-1/2"

The *QUIC-PASS®* swaging system allows "Flemish style" wire rope terminations to be swaged in only two passes. This is accomplished while maintaining currently published efficiency ratings and utilizing National Swage S-505 Standard "COLD TUFF"® Steel Sleeves.

The special design of the *QUIC-PASS®* dies allows the swaging process to be completed in just two passes, resulting in a 50-75% reduction in the number of passes required with conventional swaging systems. Unlike standard round dies, the *QUIC-PASS®* dies close completely with each pass, resulting in an increase in overall swaging process efficiencies (the job can be performed quicker), a reduction in the complexity of swaging (the concern for excess flashing between dies has been eliminated) and a reduction in training time needed for operators (more user friendly).

The finished sleeve has a "Hex" appearance that provides a *QUIC-CHECK®* look to determine if the termination has been swaged and provides a flat surface that allows for ease of I.D. stamping on the finished sleeve. Refer to page 24 of the *Wire Rope End Termination User's Manual* for proper die selection.

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Dies for S-501 & S-502 Swage Sockets

Swaging all S-501 & S-502 Swage Sockets requires the use of single cavity die. This is a special die designed with a relief for swage sockets and extra length to swage the full length of the shank. Refer to pages 36 and 37 of the *Wire Rope End Termination User's Manual* for proper die selection.

Swage Sockets for Spiral Strand Rope
Our tests indicate that if the spiral strand is 1 x 19 or greater, and the ultimate strength does not exceed Table 4 of ASTM A586, you can use dies for size swage sockets up to the 1-1/4". For sizes greater than 1-1/4" the following table will apply:

If the strand is of greater strength than Table 4 or has less metallic area, we must recalculate the design and test for adequacy.

Dies for S-506 Turnback Sleeves

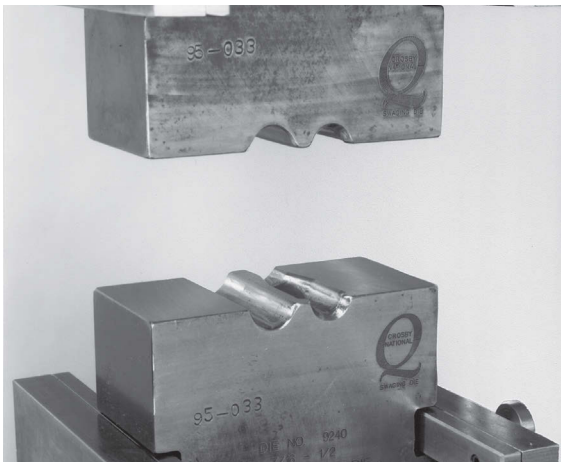
Turnback eye terminations using 5/16" through 1" S-506 Sleeves utilize the S-505 Standard Steel Sleeve die (1st Stage open channel die only). The 1-1/4" S-506 Sleeve utilizes the 1-3/8" socket (S-501 and S-502) die. Refer to page 46 of the *Wire Rope End Termination User's Manual* for proper die selection.

Dies for S-409 Buttons

Buttons are swaged in open channel dies. Refer to page 43 of the *Wire Rope End Termination User's Manual* for proper die selection.

Specific recommended swaging practices can be found in each product section of this manual. The proper die selection and the recommended maximum after swage dimensions are referenced in the section of this manual that contains the product you are swaging. This information can also be found in The Crosby General Catalog (See Section "Wire Rope End Terminations"), the National Swage Die Guide, or by referring to the National Swage Die Chart.

Dies and die adapters to fit other type swaging machines are available upon request (Refer to page 19).



Two Cavity Die



Never use dies that are cracked, worn or abraded (galled).



Single Cavity Die

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After Swage Inspection Procedures

⚠ WARNING

- **Read, understand, and follow these instructions before using the National QUIC-PASS® Swaging System.**
- **Improper after swage dimensions can result in sling failure resulting in property damage, serious injury or death.**
- **Always gauge or measure the after swage dimensions to ensure proper sling performance.**
- **Using National Swaging System with ropes and termination styles other than shown in these procedures may reduce the performance of the termination and lead to premature failure.**
- **When using rope constructions other than shown in this procedure, the termination must be destructive tested and documented to prove adequacy of the assembly to be manufactured.**
- **The QUIC-PASS® Swaging System is designed only for "Flemish Eye" terminations using National S-505 Standard Steel Sleeves.**
- **The QUIC-PASS® Swaging System is not designed for Cable-Laid wire rope slings.**

Checking Swaging Dimensions

One of the important considerations in producing a quality termination is the overall diameter of the fitting after the swaging process is complete. Since all dies wear, and the swaged fitting used in terminations have spring back, the results of swaging should be checked periodically to determine the wear condition of the die as well as to ensure the fitting is swaged to proper dimensions.

Key Facts About After Swage Dimensions:

1. In addition to worn dies, not achieving the proper after swage dimension can also be due to the die not being fully closed during swaging. Dies showing excessive wear should be replaced.
2. The effective swaging that dies can accomplish stops when the die lands touch each other. Any continued swaging adds needless wear and strain on the dies and swaging machine.
3. By placing a light oil on the die faces and in the cavity, the dies will be lubricated as well as protected.
4. The oozing of the oil from the faces of the dies as they touch will indicate when the dies have closed. At this point, stop the swaging cycle.
5. Additional swaging adds needless wear and strain to the dies and swaging machine.
6. Never use dies that are cracked, worn or abraded (galled).
7. The Crosby Group does not recommend the checking of die dimensions as an acceptable method of determining the quality of a swage sleeve, button, ferrule, or socket.
8. It is our recommendation that the checking of the after swage dimension of the swaged fitting is the most accurate indicator of a properly swaged termination. Measuring the die cavity only is not an acceptable process control check.
9. If the die cavity wears, the dies are not closed completely during swaging. If an inadequate number of presses are used, it could be quickly identified by checking the after swage dimension of the part.
10. Swaging Machine not producing sufficient tonnage will affect after swage dimensions.

No-Go Gauge Information

To assist in checking the after swage dimensions of the fitting, the Crosby Group provides the National No-Go Gauges. When used correctly the National No-Go Gauges can determine if the fittings were swaged to the proper diameter. We would recommend that all Crosby products or product swaged in Crosby dies be checked with the proper gauge to determine the acceptability of the swaging process.

- Gauges are made of hardened alloy steel and machined to strict tolerances.
- Gauge can be used to verify that all fittings have been swaged properly.
- After swage dimensions not within the maximum limits may result from worn dies or improper swaging techniques.
- Other type gauges are available upon request.
- National No-Go Gauges are available for a variety of products (See Table 1).
- **No-Go Gauges and QUIC-PASS® No-Go Gauges are not interchangeable.**

Table 1

Fitting and Size	Part No.
505 Sleeve 1/4 - 7/8	1095512
505 Sleeve 1 - 1-1/2	1095521
505 Sleeve 1-3/4	1095530
505 Sleeve 2	1095549
505 Sleeve 2-1/4	1095558
505 Sleeve 2-1/2	1095587
505 Sleeve 2-3/4	1095576
505 Sleeve 3	1095565
505 Sleeve 3-1/2	1095594
505 Sleeve 3-3/4	1095601
505 Sleeve 4	1095610
501/502 Socket 1/4 - 1	1095647
501/502 Socket 1-1/8 - 1-3/4	1095656
501/502 Socket 2	1095665

Using No-Go Gauges

When swaged properly, the gauge will go up and down (see Figure 1) and around the full length of the fitting (see Figure 2).

For the proper after swage dimensions, see the section in this publication for the specific product you are swaging.

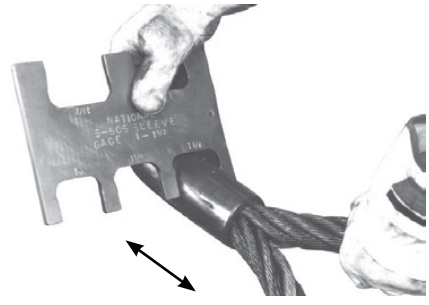


Figure 1

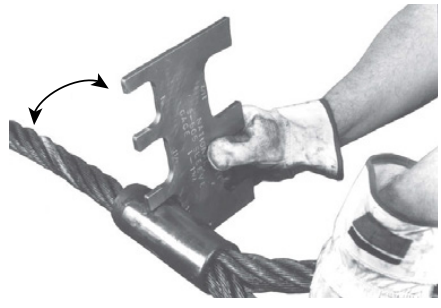


Figure 2

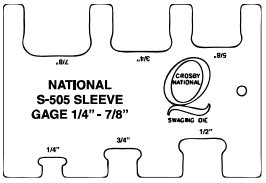
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QUIC-PASS® No-Go Gauges

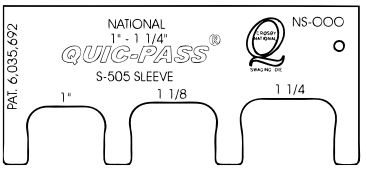
As a further aid, QUIC-PASS® No-Go gauges are available for checking the sleeve's dimensions after swaging is complete.

- Gauges are made of hardened alloy steel and machined to strict tolerances.
- Gauge can be used to verify that all sleeves have been swaged properly.
- "After Swage" dimensions not within the maximum limits may result from worn dies or improper swaging techniques.
- **No-Go Gauges and Quic-Pass® No-Go Gauges are not interchangeable.**

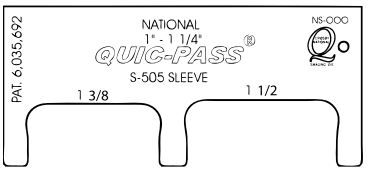
QUIC-PASS® No-Go Gauges	
Sleeve and Size	**Stock No.**
No-Go Gauge for S-505 1/4" - 7/8"	1923705
No-Go Gauge for S-505 1" - 1-1/4"	1923712
No-Go Gauge for S-505 1-3/8" - 1-1/2"	1923714



Stock No. 1923705

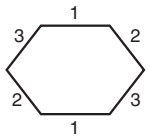


Stock No. 1923712



Stock No. 1923714

Use a National QUIC-PASS® No-Go Gauge to check the after swage dimensions to ensure that it has been swaged to the proper dimension. When swaged properly, the gauge will slide up and down the full length of the sleeve on all three sets of opposing flats.



QUIC-PASS® Maximum After Swage Dimensions

Size	Maximum "After Swage" Dimension (in.)
1/4	0.565
5/16 - 3/8	0.769
7/16 - 1/2	1.016
9/16 - 5/8	1.247
3/4	1.475
7/8	1.738
1	1.955
1-1/8	2.170
1-1/4	2.405
1-3/8	2.610
1-1/2	2.835

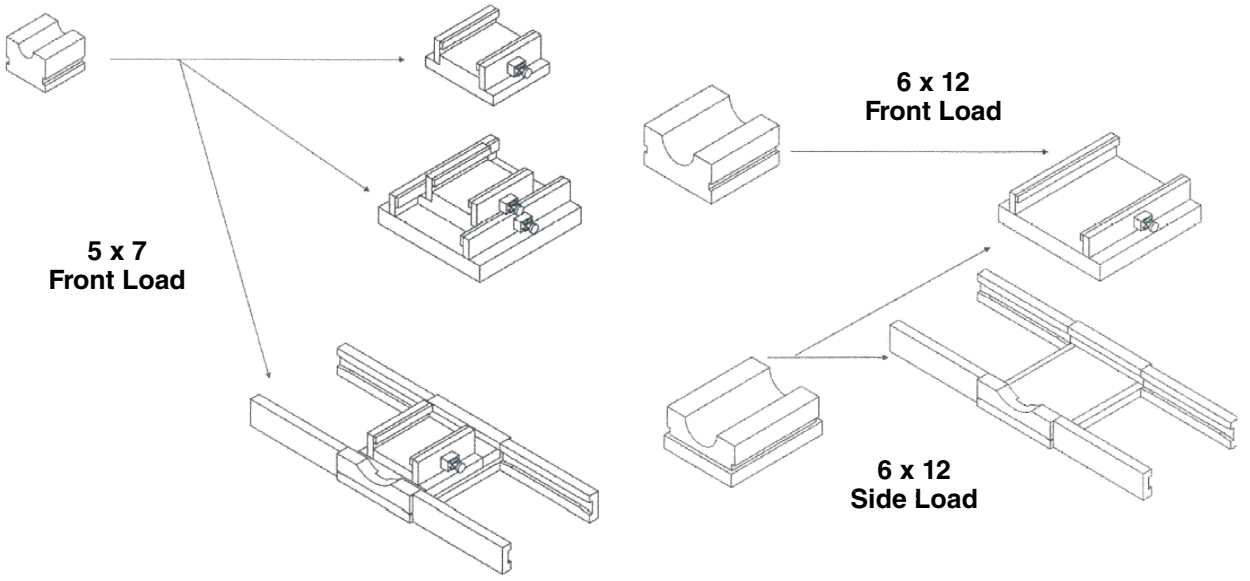
Important Safety Information

- Crosby does not recommend a "Texas Tuck" style termination with Crosby National S-505 "COLD TUFF®" Standard Steel Sleeves.
- Only Crosby National S-505 "COLD TUFF®" Standard Steel Sleeves are recommended when using the QUIC-PASS® Swaging System.
- National S-505 Standard Steel Sleeves, when used with the QUIC-PASS® Swaging System, are only recommended for use with one (1) part 6 X 19 or 6 X 37, IPS or XIP (EIP), XXIP (EEIP), RRL, IWRC rope.
- The condition of the swaging machine can cause sleeve "After Swage" size not to be within the proper dimensions. Example: worn bushings, loose tie rods, loose die holders, misaligned platens, worn pins, worn linkage, etc.
- Swaging dies being worn, damaged, misused, or undersized can cause sleeve "After Swage" size not to be within the proper dimension.
- Swaging die holders excessively worn, damaged, misused or loose can cause sleeve "After Swage" size not to be within the proper dimension. Only use QUIC-PASS® dies and die holders inspected and properly secured in National swaging machines.
- Always refer to Warning and Application information found in the *Crosby General Catalog* and *Wire Rope End Terminations User's Manual*

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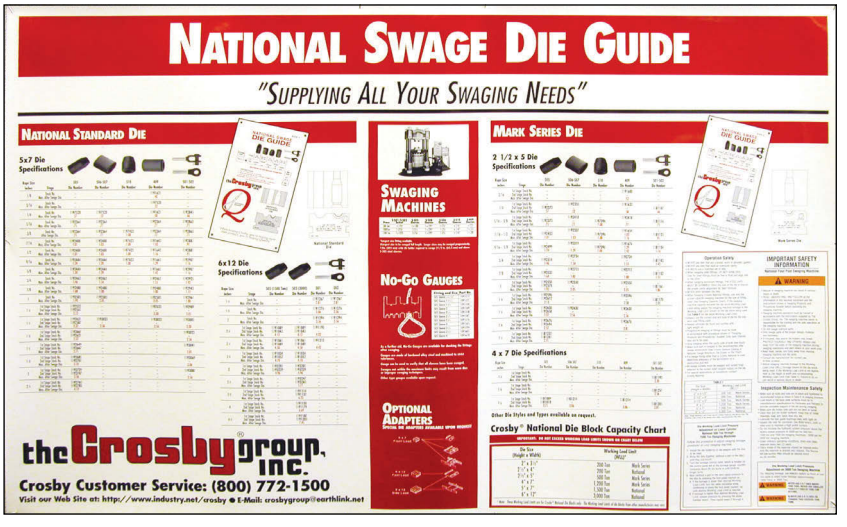
Die Information

Optional Adapters



* Special dies and adapters available upon request.

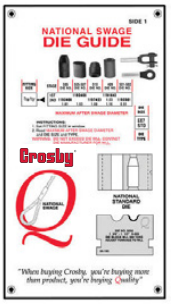
Proper die selection and after swage dimensional information can also be found in the following Crosby® sources.



National Swage Die Guide Chart



Crosby
General Catalog



National
Swage
Die Guide

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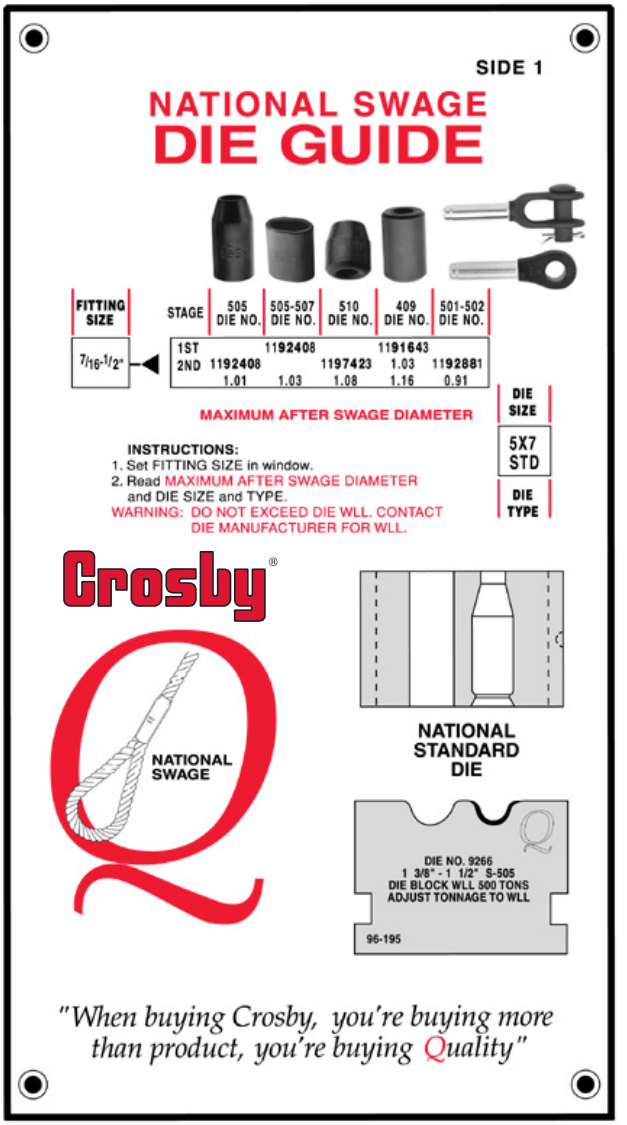
Die Information

The National Swage Die Guide was developed to aid in determining which dies are required for swaging the various sizes and types of products manufactured by National Swage.

Side 1 of the Die Guide is to be referenced when working with National Standard Dies.

Side 2 of the Die Guide is to be referenced when using Mark Series Dies.

IMPORTANT: Be sure to reference the correct information (side 1 or side 2) for your particular swaging application.



USING THE NATIONAL SWAGE DIE GUIDE

An example of how to use the Die Guide would be if you needed to swage a 1/2" S-501 Swage Socket in National Standard Dies. First, find the 7/16 -1/2" fitting size designation, and follow that row across the different product types to the column heading 501 - 502 Die No.

You can see that the required Die No. is 1192881 and the size and type is 5 x 7 Std. When you have located and verified that you have the correct dies, you are ready to begin the swaging process. You can also find the maximum after swage dimension listed below the corresponding Die No. (i.e. 0.91)

Remember, side 1 refers to National Standard Dies and side 2 refers to Mark Series Dies.

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COLD TUFF® Swage Fittings – Introduction



A number of years ago, we met the challenge of making swage fittings, including sleeves, sockets and buttons more reliable during adverse swaging conditions. At that time sleeves tended to crack during the swaging process – especially at lower temperatures.

Our research in materials, and the development of a unique thermal process resulted in the development of COLD TUFF® swage fittings. This name was derived from the ability of these new materials to cold flow plastically (a definition of swaging) in the presence of a notch at extremely low temperatures.

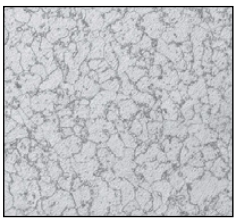
The success of COLD TUFF® swage fittings in overcoming problems encountered in swaging has revolutionized the industry. The result of the COLD TUFF® process has made National Swage fittings stronger. National S-505 Standard Steel sleeves have outperformed stainless steel sleeves on wire rope eye terminations in the classic 90 degree angle between the rope legs of sling eye test – and for a fraction of the cost of a stainless steel sleeve.

Also, COLD TUFF® swage products can be impression stamped after swaging for identification purposes. Properly done, there should be no concern for fractures initiating from such impressions. Specific recommended stamping procedures can be found in each product section.

Key Facts About Swaging:

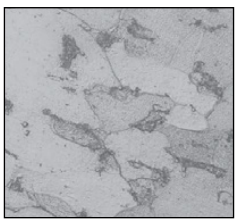
1. Crosby's proprietary COLD TUFF® heat treat process has made swaging more reliable during the demanding conditions of swaging.
2. COLD TUFF® virtually eliminates the cracking during the swaging process, especially at extremely low temperatures.
3. The steel microstructure created from the COLD TUFF® heat treatment process reveals a uniform structure in the Crosby fittings that result in Charpy impact values averaging up to ten times greater than those of competitive products.

Un-retouched Photomicrographs of Sleeve (Original Magnification @ 500x)



Crosby COLD TUFF® Heat Treatment

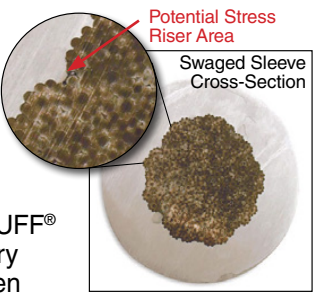
The ***uniform*** microstructure of the COLD TUFF® process results in superior ductility and toughness properties.



Competitor's Heat Treatment

The non-uniform microstructure may lack proper ductility and toughness required for swaging and use.

4. The improved steel microstructure provides increased resistance to cracks that can occur during the sling building process. Cracks are initiated at stress risers.
5. Crosby COLD TUFF® sleeves dramatically reduce the effects of stress risers that can occur during swaging where the "cold worked" sleeve makes contact with the wire. Stress risers are not detectable through visual inspection.
6. Because of the higher Charpy values found in Crosby COLD TUFF® sleeves, stress risers caused by stenciling are not detrimental.
7. Crosby or CG and COLD TUFF® are clearly stenciled on every sleeve we make (legible even after swaging) leaving no question as to the quality or the manufacturer.



COLD TUFF® process results in superior ductility that reduces the effects of stress risers.

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COLD TUFF® Swage Fittings – S-505

Swaging Methods for S-505 Sleeves

(Carbon Steel Sleeve for Flemish Eyes)

National S-505 carbon steel sleeves are specially designed for use with Flemish Eyes and for swaging in National dies.

These swage fittings are designed to be used on one (1) part of 6 x 19 or 6 x 37 FC IPS or IWRC EIPS Right Regular Lay wire rope.

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.

Cold Swaging Process

Cold swaging is a process which causes considerable movement and work hardening of the steel in the fitting. Exerting extreme pressures, the swaging machine forces the steel to flow into the crevices between the wires and strands of the wire rope. At the same time, the swaging machine forces the material to flow plastically along the length of the wire rope. These are a few of the reasons swaging is done on steel fittings in multiple passes in a well lubricated die.

Protecting Dies and Swaging Machines

When the die lands touch each other, effective swaging stops. Additional swaging adds needless wear and strain on the dies and swaging machine. We recommend placing oil on the die faces and cavities. When swaging, the oozing of the oil from the die faces will indicate when the dies are closed. At this point, stop swaging.

Avoid Excess Flashing

Flashing occurs when sleeve material flows out into the area between the die faces. Once formed, additional passes will not eliminate flashing. They will only fold back the flash and develop a permanent mark in the material and possibly create a crack.

Flashing is not an issue with QUIC-PASS® system, as each pass requires full closure of the dies.

To avoid excess flashing, do the following:

- Select proper dies.
- Make sure that dies are in good condition and properly lubricated.
- Apply lubricant such as a light hydraulic oil to both die cavities and the lower die face.
- Follow swaging instructions on pages 27-35.



CORRECT **EXCESS FLASHING**



Identification Stamping

For identification purposes, National COLD TUFF® swage products can be impression stamped after swaging. Properly done, there should be no concern for fractures initiating from such impressions.



To stamp S-505 Sleeves:

- Use round corner or low stress stamps. Stamp to maximum depth of 0.015 in. (1/64”).
- Stamp in area on the side of the sleeve in the plane of the sling eye, and no less than 0.250 in. (1/4”) from either end of the sleeve.

NOTE:

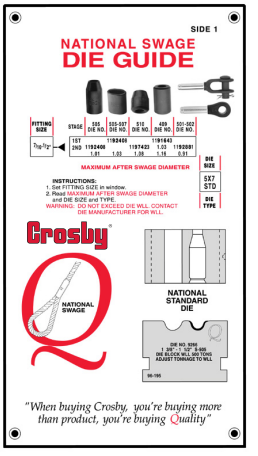
WHEN USING NATIONAL QUIC-PASS® DIES, THE FINISHED SLEEVE HAS A “HEX” APPEARANCE THAT PROVIDES A QUIC-CHECK® LOOK TO DETERMINE IF THE TERMINATION HAS BEEN SWAGED, AND PROVIDES A FLAT SURFACE THAT ALLOWS FOR EASE OF I.D. STAMPING ON THE FINISHED SLEEVE.

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National Steel Swaging Sleeves



S-505



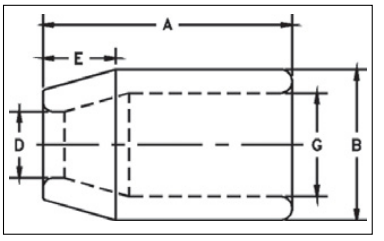
The National Die Guide will assist you in selecting the proper dies to meet your swaging needs.

- For Flemish Eye wire rope splicing.
- Designed for low temperature toughness.
- Resists cracking when swaged (equals or exceeds stainless steel sleeves).
- Special processed low carbon steel.
- "COLD TUFF®" for better swageability.
- Can be stamped for identification after swaging without concern for fractures when following these directions.
  - Use round corner stamps to a maximum depth of 0.015 in. (1/64"). The area for stamping should be on the side of the sleeve in the plane of the sling eye, and no less than 0.250 in. (1/4") from either end of the sleeve.
- Standard Steel Sleeve terminations have efficiency ratings as follows based on the catalog strength of wire rope.

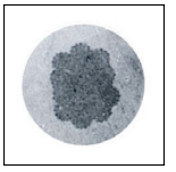
S-505 Termination Efficiency		
Size (in.)	Type of Wire Rope*	
	IWRC	FC
1/4 - 1	96%	93%
1-1/8 - 2	92%	89%
2-1/8 and Larger	90%	87%

*** NOTE: S-505 Standard Sleeves are recommended for use with 6 x 19 or 6 x 37, IPS or XIP (EIP), XXIP (EEIP), RRL, FC or IWRC wire rope.**

Before using any National Swage fitting with any other type lay, construction or grade of wire rope, it is recommended that the termination be destructive tested and documented to prove the adequacy of the assembly to be manufactured.



NOTE: See page 24 for dimensional information.



"Cross Section of Swaged Sleeve"

QUIC-PASS®

National QUIC-PASS® Swaging System

"The Next Generation in Swaging Systems"

The ***QUIC-PASS®*** swaging system allows "Flemish style" wire rope terminations to be swaged in only two passes. This is accomplished while maintaining currently published efficiency ratings and utilizing National Swage S-505 Standard "COLD TUFF®" Steel Sleeves.

- Allows the swaging process to be completed in just two passes, resulting in a 50-75% reduction in the number of passes required with conventional swaging systems.
- Allows the dies to close completely with each pass, resulting in...
  - *An increase in overall swaging process efficiencies* (the job can be performed quicker).
  - *A reduction in the complexity of swaging* (the concern for excess flashing between dies has been eliminated).
  - *A reduction in training time needed for operators* (more user friendly).
- The finished sleeve has a *"Hex"* appearance that provides a ***QUIC-CHECK®*** look to determine if the termination has been swaged and provides a flat surface that allows for *ease of I.D. stamping on the finished sleeve.*

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